

MONIL ARORA

Vancouver, BC

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Education

University of British Columbia, Vancouver

Expected 2027

Bachelor of Applied Science in Electrical Engineering

Relevant Coursework: Digital System Design, Computing Systems, Signals and Systems, Systems and Controls

Projects

Self-Balancing Robot | C, Arduino, ESP, Sensors, Embedded Systems

Jan – Apr 2025

- Designed and built a two-wheeled **self-balancing robot** using **Arduino Nano 33**, **IMU sensors**, and **motor drivers**, implementing real-time **PWM control** for tilt stabilization up to **45°**.
- Developed a **BLE-enabled mobile app** and **LCD remote** using **ESP32** to transmit commands and display RPM, tilt angle, and sonar distance readings.
- Validated proportional control through **oscilloscope analysis**, achieving approximately **30% improvement in stability**.

Deep Neural Network Accelerator | Verilog, Assembly, C, FPGA

Oct – Nov 2024

- Implemented a **deep neural network accelerator** on an embedded **Nios II system** using the **DE1-SoC board** for MNIST classification with **Q16.16 fixed-point arithmetic** and **ReLU** activation.
- Developed a **system-on-chip (SoC)** featuring an **SDRAM interface**, applying **Quartus timing constraints** to ensure stable operation at **50 MHz**.

Metal Detector Robot | C, STM32, EFM8, JDY40 Bluetooth

Mar – Apr 2024

- Collaborated with a team of six to design a battery-powered, **remote-controlled metal-detection robot** using **STM32** and **EFM8 microcontrollers**.
- Programmed robust **communication protocols** and an **LCD interface** for real-time visualization of **metal detection intensity** and **magnetic field strength (up to 1.4 T)**.

Reflow Oven Controller | Assembly, ARM

Jan – Feb 2024

- Developed a **microcontroller-based reflow oven controller** using the **N76E00 (ARM architecture)** in **assembly language** to automate **soldering temperature profiles** ranging from **25°C to 240°C**.
- Achieved precision within **±3°C**, integrating **automatic safety shutdowns and safety alerts**, a **real-time PC monitoring interface**, and support for multi-stage temperature ramping.

RISC Machine | Verilog, ARM, FPGA

Oct – Dec 2023

- Designed and implemented a **CPU** with integrated **datapath** and **FSM controller** on the **DE1-SoC board**, integrating **synchronous RAM** for efficient instruction and data handling.
- Extended the instruction set with **load, store, and halt operations** to enhance control flow and execution flexibility.

Work Experience

Sharang Shakti Pvt Ltd

May – Jun 2024

Embedded Signal Processing Engineer

Gurugram, India

- Contributed to the **digitalization of an analog radar system** by designing and optimizing high-frequency waveform generation and signal conditioning modules.
- Enhanced **AD9910 DDS-based waveform generation** to improve signal resolution and reduce phase noise, enabling precise frequency control for radar testing.
- Integrated the **AD8336 Variable Gain Amplifier (VGA)** with the **AD9910** for real-time amplitude modulation and dynamic range adjustment in **high-frequency radar signals**.

Technical Skills

Languages: C, C++, Assembly, Verilog, VHDL, ARM

Tools: Quartus, ModelSim, CrossIDE, Cadence, MATLAB, LTspice, Git, VS Code, Visual Studio, SolidWorks

Hardware: DE1-SoC, ESP32, STM32, EFM8, Arduino, Raspberry Pi, IMU Sensors, Motor Drivers